

FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-114 Data Engineering Job Transition Guide

Evidence-first portfolio, resume, interviews and role targeting after graduation

Status	RC1 - FINAL-LAYER QA PASS CANDIDATE
Course boundary	C3-104 = C3-09 phase Gate only; C3-113 = separate course-wide graduation Gate
Release boundary	Course 3 is NOT FINAL until Mentor/PWA + whole-course release regression are complete.

Permanent engineering behavior controls the course. Vendor UI/version notes are refreshable references, and external links are support-only; the course artifacts remain sufficient without them.

1. What Course 3 proves - and what it does not

Use Course 3 to prove structured engineering capability. Do not represent synthetic projects as production employment experience. Strong evidence can justify junior/associate interviews; employer experience remains a separate claim.

2. Role targeting

Junior / Associate Data Engineer	Primary external target when SQL/Python/pipelines + capstone evidence are strong.
Data Engineer I	Apply where scope matches demonstrated independence and operational depth.
Analytics Engineer	Strong bridge when SQL/dbt/modeling/warehouse work dominates.
Internal DE transfer	Especially valuable after analyst experience because company/domain context transfers.

3. Portfolio package

- Lead with RetailPulse as one coherent production-style system, not many disconnected notebooks.
- Show ADR/architecture, source contract, incremental logic, tests, reconciliation, orchestration, Spark proof, quality/observability, security boundary, CI/change process, incident recovery and runbook.
- Include a 10x-scale discussion and explain why streaming is or is not justified.
- Sanitize screenshots/logs; remove secrets, local machine paths and private data.
- Keep an evidence index mapping each resume/interview claim to a runnable artifact.

4. Resume bullet formula

Action + system + engineering control + verifiable technical evidence. Avoid invented business impact.

Example: Built a synthetic commerce data platform with incremental order corrections, reproducible orchestration, PySpark transformation, blocking quality checks, source-to-serving reconciliation, automated validation and documented incident recovery.

5. Interview defense map

SQL / modeling	grain, keys, joins/windows, history, warehouse/lakehouse trade-offs, reconciliation
Python	modules, config, exceptions/logging, tests, file/API/database ingestion
Pipelines	idempotence, retries, backfills, partial failure, late data, state/checkpoints
Spark	partitions, shuffles, joins, skew, plans, testability, when Spark is unnecessary
Cloud / Fabric	Lakehouse/Warehouse, pipelines/notebooks, OneLake/security, monitoring/lifecycle; portable vs vendor-specific
Operations	quality gates, SLOs, incidents, recovery, rollback/roll-forward, evidence

6. Application rule

Match demonstrated evidence to each job description. Do not reject AWS/GCP roles solely because the course uses Fabric: permanent engineering patterns transfer, while vendor services can be learned. Always distinguish what you actually executed from what you only understand conceptually.